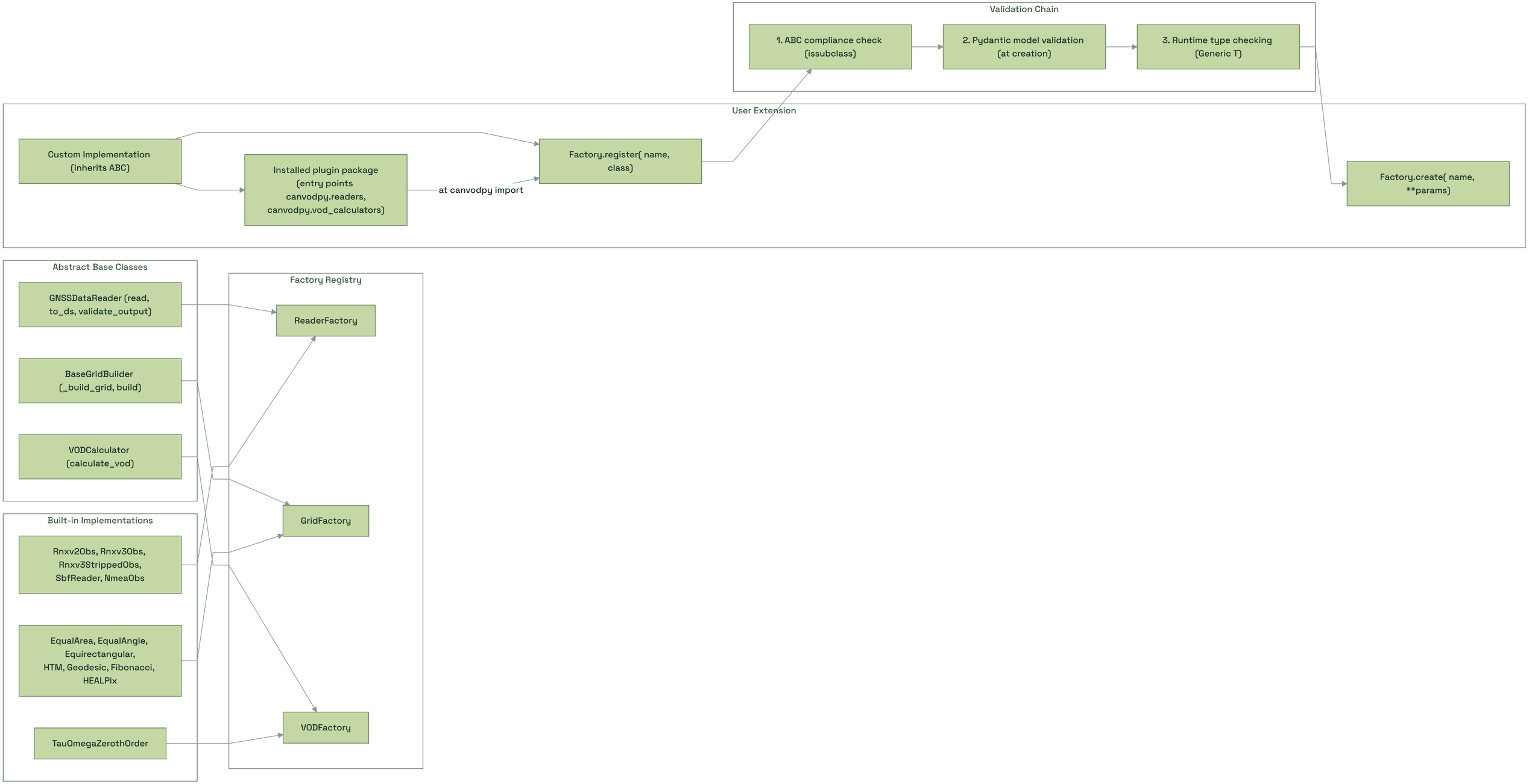
Validation Chain
1. ABC compliance check (issubclass)
2. Pydantic model validation (at creation)
3. Runtime type checking (Generic T)
User Extension
Custom Implementation (inherits ABC)
Installed plugin package (entry points canvodpy.readers, canvodpy.vod_calculators)
at canvodpy import
Factory.register(name, class)
Factory.create(name, **params)
Abstract Base Classes
GNSSDataReader (read, to_ds, validate_output)
BaseGridBuilder (_build_grid, build)
VODCalculator (calculate_vod)
Factory Registry
ReaderFactory
GridFactory
VODFactory
Built-in Implementations
Rnxv2Obs, Rnxv3Obs, Rnxv3StrippedObs, SbfReader, NmeaObs
EqualArea, EqualAngle, Equirectangular, HTM, Geodesic, Fibonacci, HEALPix
TauOmegaZerothOrder